

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

Student Name: R.Mohini

Roll No.: A24126552110

1. TITLE

Design Webpage Using HTML5 & CSS3

2. AIM

To design and develop a responsive and visually appealing webpage using HTML5 structural elements and CSS3 styling properties, and to understand how HTML provides the structure of a webpage while CSS3 is used to control its presentation, layout, colors, fonts, spacing, and overall appearance.

3. OBJECTIVE

- To understand the basic structure of an HTML5 document.
- To use HTML5 semantic elements such as <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer>.
- To create a well-structured webpage using HTML5 elements.
- To understand the role of CSS3 in styling HTML elements.
- To apply CSS3 selectors such as element, class, and ID selectors.
- To use CSS properties for colors, fonts, margins, padding, borders, and backgrounds.
- To design webpage layouts using CSS3 Flexbox and/or Grid.
- To create a responsive webpage that adapts to different screen sizes.
- To understand the separation of content from presentation using HTML and CSS.
- To improve the visual appearance and usability of a webpage using CSS3.

4. THEORY

HTML5 (HyperText Markup Language) is the standard markup language used to create and structure content on webpages. It provides the basic structure of a webpage using elements such as headings, paragraphs, links, images, lists, forms, tables, and semantic elements.

HTML5 introduced semantic elements such as <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer>. These elements clearly describe the purpose of different parts of a webpage and make the document easier to understand, maintain, and access.

CSS3 (Cascading Style Sheets) is used to control the presentation and appearance of HTML elements. CSS3 can be used to define colors, fonts, backgrounds, borders, spacing, alignment, dimensions, animations, transitions, and layouts.

CSS can be applied to HTML using three methods: Inline CSS, Internal CSS, and External CSS. Inline CSS is written inside an HTML element, Internal CSS is written inside a <style> element, and External CSS is written in a separate .css file and linked to the HTML document.

CSS3 provides modern layout techniques such as Flexbox and CSS Grid. Flexbox is useful for arranging elements in a row or column, while Grid is useful for creating two-dimensional layouts.

CSS3 also supports media queries, which allow a webpage to change its layout and styling according to the screen size. This helps in creating responsive webpages that work properly on desktops, tablets, and mobile devices.

Thus, HTML5 is mainly responsible for the structure and content, whereas CSS3 is responsible for the design and presentation of the webpage.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare <!DOCTYPE html> at the beginning.
2. Define the <html>, <head>, and <body> sections.
3. Add the character encoding using <meta charset="UTF-8">.
4. Add a suitable title using the <title> element.
5. Create the webpage structure using HTML5 semantic elements such as <header>, <nav>, <main>, <section>, <article>, <aside>, and <footer>.
6. Add headings, paragraphs, links, lists, images, and other required HTML elements.
7. Create a CSS section or a separate CSS file for styling the webpage.
8. Apply CSS selectors to target the required HTML elements.
9. Apply colors, fonts, backgrounds, borders, margins, padding, and other styling properties.
10. Use Flexbox or CSS Grid to arrange the webpage components properly.
11. Apply hover effects to interactive elements such as navigation links and buttons.
12. Use media queries to make the webpage responsive to different screen sizes.
13. Save the HTML file and CSS file with appropriate extensions.
14. Open the HTML file in a web browser.
15. Verify the webpage structure, styling, alignment, navigation, and responsiveness.
16. Test the webpage at different screen sizes and record the observations.

6. PROGRAM

```
<!DOCTYPE html>
```

```
<html lang="en">
```

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>HTML5 & CSS3 Webpage</title>

<style>

/* General Styling */

* {

margin: 0;

padding: 0;

box-sizing: border-box;

}

body {

font-family: Arial, sans-serif;

background: linear-gradient(135deg, #dbeafe, #e9d5ff);

color: #333;

line-height: 1.6;

}

/* Header */

header {

background: linear-gradient(135deg, #4f46e5, #7c3aed);

color: white;

padding: 30px;

text-align: center;

}

```
header h1 {  
    margin-bottom: 10px;  
    font-size: 35px;  
}  
  
/* Navigation */  
  
nav {  
    background-color: #312e81;  
    text-align: center;  
    padding: 15px;  
}  
  
nav a {  
    color: white;  
    text-decoration: none;  
    margin: 0 20px;  
    font-weight: bold;  
}  
  
nav a:hover {  
    color: #facc15;  
}  
  
/* Main Container */  
  
main {  
    width: 90%;  
    margin: 25px auto;
```

```
}  
  
/* Hero Section */  
  
.hero {  
    background-color: white;  
    padding: 45px;  
    text-align: center;  
    border-radius: 15px;  
    margin-bottom: 25px;  
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.15);  
}  
  
.hero h2 {  
    color: #4f46e5;  
    margin-bottom: 10px;  
    font-size: 28px;  
}  
  
.hero p {  
    margin-bottom: 20px;  
}  
  
.btn {  
    display: inline-block;  
    background-color: #4f46e5;  
    color: white;  
    padding: 12px 25px;  
    text-decoration: none;
```

```
        border-radius: 6px;
    }

    .btn:hover {
        background-color: #7c3aed;
    }

    /* Content Layout */

    .container {
        display: flex;
        gap: 25px;
    }

    /* Main Articles */

    .content {
        flex: 3;
    }

    article {
        background-color: white;
        padding: 25px;
        margin-bottom: 25px;
        border-radius: 12px;
        box-shadow: 0 4px 10px rgba(0, 0, 0, 0.12);
    }

    article h3 {
        margin-bottom: 12px;
        color: #4f46e5;
```

```
}

/* Sidebar */

aside {
    flex: 1;

    background-color: white;

    padding: 25px;

    border-radius: 12px;

    height: fit-content;

    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.12);
}

aside h3 {
    margin-bottom: 12px;

    color: #7c3aed;
}

aside ul,
aside ol {
    padding-left: 20px;
}

aside li {
    margin-bottom: 8px;
}

/* Form */

form {
    margin-top: 15px;
```

```
}
```

```
label {
```

```
    display: block;
```

```
    margin-top: 12px;
```

```
    margin-bottom: 5px;
```

```
    font-weight: bold;
```

```
}
```

```
input,
```

```
select,
```

```
textarea {
```

```
    width: 100%;
```

```
    padding: 11px;
```

```
    border: 1px solid #aaa;
```

```
    border-radius: 6px;
```

```
    font-size: 14px;
```

```
}
```

```
input:focus,
```

```
select:focus,
```

```
textarea:focus {
```

```
    outline: none;
```

```
    border-color: #4f46e5;
```

```
}
```



```
/* Buttons */
```

```
button {  
    margin-top: 15px;  
    margin-right: 8px;  
    padding: 11px 22px;  
    border: none;  
    background-color: #4f46e5;  
    color: white;  
    border-radius: 6px;  
    cursor: pointer;  
}
```

```
button:hover {  
    background-color: #7c3aed;  
}
```

```
/* Footer */
```

```
footer {  
    background-color: #312e81;  
    color: white;  
    text-align: center;  
    padding: 18px;  
    margin-top: 25px;  
}
```

```
/* Responsive Design */

@media (max-width: 768px) {

  .container {

    flex-direction: column;

  }

  nav a {

    display: block;

    margin: 10px;

  }

  main {

    width: 95%;

  }

  .hero {

    padding: 30px 20px;

  }

  header h1 {

    font-size: 28px;

  }

}

</style>

</head>

<body>

  <!-- Header -->
```

```
<header>

  <h1>WebDesign</h1>

  <p>HTML5 & CSS3 Webpage</p>
</header>

<!-- Navigation -->

<nav>

  <a href="#home">Home</a>

  <a href="#articles">Articles</a>

  <a href="#about">About</a>

  <a href="#contact">Contact</a>

</nav>

<!-- Main Content -->

<main>

  <!-- Hero Section -->

  <section class="hero" id="home">

    <h2>Welcome to My Webpage</h2>

    <p>

      This webpage is designed using HTML5 semantic
      elements and CSS3 styling properties.

    </p>

    <a href="#articles" class="btn">

      Explore More

    </a>
```

</section>

<!-- Two Column Layout -->

<div class="container">

<!-- Main Content -->

<section class="content" id="articles">

<!-- HTML5 Article -->

<article>

<h3>HTML5</h3>

<p>

HTML5 is used to create the structure and content of a webpage. It provides semantic elements such as header, nav, section, article, aside and footer.

</p>

</article>

<!-- CSS3 Article -->

<article>

<h3>CSS3</h3>

<p>

CSS3 is used to style HTML elements. It allows us to add colors, fonts, spacing, borders, shadows, layouts and responsive design to webpages.

</p>

```
</article>
```

```
<!-- Contact Form -->
```

```
<article id="contact">
```

```
  <h3>Contact Us</h3>
```

```
  <form>
```

```
    <label for="name">
```

```
      Name:
```

```
    </label>
```

```
    <input
```

```
      type="text"
```

```
      id="name"
```

```
      name="name"
```

```
      placeholder="Enter your name"
```

```
      required
```

```
    >
```

```
    <label for="email">
```

```
      Email:
```

```
    </label>
```

```
    <input
```

```
      type="email"
```

```
      id="email"
```

```
      name="email"
```

```
      placeholder="Enter your email"
```

```
      required
```

```
>

<label for="topic">

    Topic:

</label>

<select id="topic" name="topic">

    <option value="html">

        HTML5

    </option>

    <option value="css">

        CSS3

    </option>

    <option value="web">

        Web Development

    </option>

</select>

<label for="message">

    Message:

</label>

<textarea

    id="message"

    name="message"

    rows="5"

    placeholder="Enter your message"

></textarea>
```

```
<button type="submit">
```

Submit

```
</button>
```

```
<button type="reset">
```

Reset

```
</button>
```

```
</form>
```

```
</article>
```

```
</section>
```

```
<!-- Sidebar -->
```

```
<aside id="about">
```

```
<h3>Web Technologies</h3>
```

```
<ul>
```

```
<li>HTML5</li>
```

```
<li>CSS3</li>
```

```
<li>JavaScript</li>
```

```
<li>Bootstrap</li>
```

```
</ul>
```

```
<br>
```

```
<h3>Benefits</h3>
```

```
<ol>
```

```
<li>Easy to learn</li>
```

```
<li>Responsive design</li>
```

Better webpage structure

Attractive user interface

</aside>

</div>

</main>

<!-- Footer -->

<footer>

<p>

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</p>

</footer>

</body>

</html>

7. CODE EXPLANATION

Tag / Element / Property	Purpose
<!DOCTYPE html>	Declares the document as an HTML5 document.
<html>	Defines the root element of the HTML document.
<head>	Contains metadata, title, and other information about the webpage.
<meta charset="UTF-8">	Specifies the character encoding of the document.
<title>	Defines the title displayed in the browser tab.
<body>	Contains all visible webpage content.
<header>	Defines the top section of the webpage.
<nav>	Contains navigation links used to move between sections or pages.
<main>	Represents the primary content of the webpage.
<section>	Groups related content into a separate section.
<article>	Represents an independent piece of content.
<aside>	Contains supplementary or related content.
<footer>	Defines the bottom section of the webpage.

<h1>, <h2>, <h3>	Define headings of different levels.
<p>	Defines a paragraph.
<a>	Creates a hyperlink.
	Displays an image on the webpage.
/	Creates unordered and ordered lists.
<div>	Groups HTML elements for structure and styling.
class	Assigns a reusable name to HTML elements for CSS styling.
id	Assigns a unique identifier to an HTML element.
CSS selector	Selects HTML elements to which styles are applied.
color	Changes the text color.
background-color	Sets the background color.
font-size	Defines the size of the text.
font-family	Specifies the font used for text.
margin	Creates space outside an element.
padding	Creates space inside an element.
border	Adds a border around an element.
display: flex	Creates a flexible one-dimensional layout.
display: grid	Creates a two-dimensional grid layout.
text-align	Controls the horizontal alignment of text.
border-radius	Creates rounded corners.
box-shadow	Adds a shadow effect around an element.
:hover	Applies styling when the user places the mouse over an element.
@media	Defines responsive styles for different screen sizes.

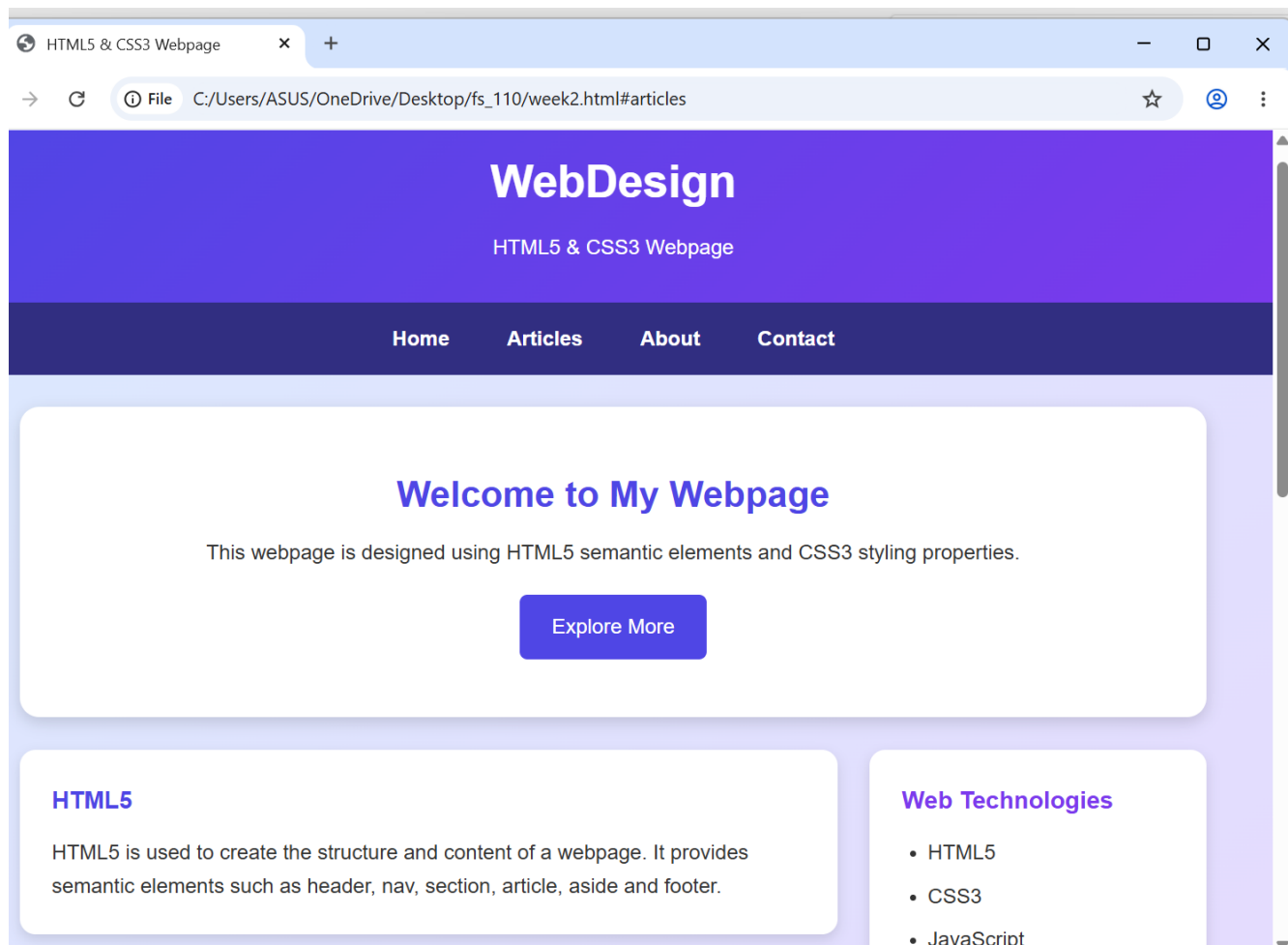
8. TEST CASES AND EXECUTION DATA

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open exp1.html in a web browser	Webpage should load successfully	Webpage loaded successfully	Pass
TC-02	Check the header and navigation bar	Header and navigation links should be displayed properly	Header and navigation displayed correctly	Pass
TC-03	Check the main webpage content	All headings, paragraphs, images, and sections should be visible	Main content displayed correctly	Pass
TC-04	Check CSS styling	Colors, fonts, spacing, borders, and backgrounds should be applied	CSS styles applied correctly	Pass
TC-05	Move the mouse	Hover effect	Hover effect	Pass

	over navigation links/buttons	should be displayed	displayed correctly	
TC-06	Resize the browser window	Webpage layout should adjust according to screen size	Layout adjusted successfully	Pass
TC-07	Open webpage on a mobile-sized screen	Content should remain readable without improper overlapping	Webpage displayed responsively	Pass
TC-08	Check footer section	Footer should appear at the bottom of the webpage	Footer displayed correctly	Pass

9. OUTPUT

Output 1:



Output 2:.

HTML5

HTML5 is used to create the structure and content of a webpage. It provides semantic elements such as header, nav, section, article, aside and footer.

CSS3

CSS3 is used to style HTML elements. It allows us to add colors, fonts, spacing, borders, shadows, layouts and responsive design to webpages.

Contact Us

Web Technologies

- HTML5
- CSS3
- JavaScript
- Bootstrap

Benefits

1. Easy to learn
2. Responsive design
3. Better webpage structure
4. Attractive user interface

Output 3:.

Contact Us

Name:

Email:

Topic:

Message:

Output 4:

Contact Us

Name:

mamtha

Email:

mamatha@gmail.com

Topic:

CSS3

Message:

i want to learn html

Submit

Reset

10. RESULT

Thus, a webpage was successfully designed and developed using HTML5 and CSS3. HTML5 semantic elements were used to create the structure and content of the webpage, while CSS3 properties were used to improve its appearance, layout, spacing, colors, and typography. Responsive design techniques were also applied to ensure that the webpage adapts properly to different screen sizes. The webpage was successfully executed and tested in a web browser, and all the test cases were verified and found to work correctly.